Zewail City
of Science, Technology and Innovation

CSAI 498 / CSAI 499
Graduation Project Proposal

Project Title:

AI Football Analysis System

Team Members:

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Semester / Year:

Fall / 2025

Date of Submission:

October 25, 2025

Abstract:

This project addresses the problem of limited access to real-time, detailed performance analytics in football. Current analysis methods are often manual, delayed, or accessible only to elite teams with expensive tools. Our proposed solution is an AI-powered system that utilizes computer vision and machine learning to analyze player movements, actions, and performance metrics in real-time. The system consists of two main platforms: a mobile app for professionals (coaches, scouts, and players) and a website for fans. The mobile app provides live match statistics such as distance covered, passes, goals, and positioning, while the website displays post-match reports and comparative player analytics. The expected outcome is a system that enhances tactical decision-making, improves player development, and deepens fan engagement with the game.

1. Problem Statement and Motivation

Football is one of the most popular sports worldwide, yet access to in-depth performance analytics remains limited to high-budget clubs. Coaches and scouts often lack real-time data to make informed tactical decisions during matches. Players do not have easy tools to track their physical and technical performance over time. Fans, on the other hand, have minimal access to analytical insights that can enrich their understanding of the game. This limitation creates a gap between professional-level analysis and everyday football experiences. By providing accessible, automated, and real-time analytics, we can democratize football data, making advanced insights available to all stakeholders—coaches, players, and fans alike. The motivation behind this project is to bridge this gap using affordable AI technology, improving player performance, enhancing tactical decisions, and increasing audience engagement

2. Proposed Solution

- **Core Idea and Approach:**
  Our solution is a **mobile app** and **website** that provides real-time player and match analysis in soccer. The mobile app is designed for professionals like coaches, scouts, and players to access live match stats during the game, using **computer vision** technology to track player movements and key events. The app will show stats like goals, assists, shots, and distance covered directly on the match stream. After each match, it will also generate a summary of each player's performance. The **website** will provide fans with **post-match stats** and player summaries, allowing them to stay connected with the game even after it ends.

- ## Innovation and Distinctiveness

  What makes this solution unique is the combination of **real-time data** during matches with historical player performance tracking. By using computer vision, the app can track player movements and detect key events (goals, assists, etc.) in real-time, displaying the stats directly on the match stream, much like in professional broadcasts. Most existing platforms only offer post-match statistics or slow, static data, but our system updates stats live, making the game more interactive and engaging for both professionals and fans.

- ## Key Features and Functionalities

  1. **Real-Time Match Analysis (App):**

     Live Player Stats: Shows stats like goals, assists, distance covered, shots, etc., in real-time on the match stream.

     Computer Vision Tracking: Tracks players and key events (like goals) during the game, updating stats live.

     Interactive Player Profiles: Coaches, scouts, and players can access a player's detailed profile, including real-time and historical performance data.

     Match Event Detection: Automatically detects key events like goals, assists, fouls, and displays these in real-time on the app.

  2. **Post-Match Stats (Website)**

     Detailed Stats: After the game, the website shows detailed player and match stats, including possession, shots on target, and performance metrics.

  Player Summaries: Provides a summary of each player's stats from the last 5 matches(for example), helping coaches and players track progress.

  3. **Historical Performance Tracking:**

     Automatically generates and stores summaries of player stats after each match, allowing professionals to easily track player performance over time.

  The combination of live analysis, real-time stats, and historical performance tracking offers a unique, interactive experience that helps improve decision-making for professionals and enhances fan engagement.

3. Project Scope

In Scope

Must-Have Features

- Real-time Match Analysis:
- Live player stats displayed on the match stream (goals, assists, shots, distance covered, etc.).
- Computer vision tracking to follow player movements and detect key events (goals, assists, fouls).
- Match event detection (e.g., goals, assists, fouls) displayed instantly.

Post-Match Stats:

- Detailed player stats available after the match on the website (e.g., goals, assists, pass accuracy, etc.).
- Player summaries for the last 5 matches for EX (In the App), showing historical performance.

Player Profiles:

- Accessible profiles for each player, including real-time stats during matches and performance summaries after the game. (In the App)
- Display stats and analysis After the Match, similar to La Liga broadcasts but after the match.

Chatbot Integration:

- Q&A

Could-have Features:

Social Sharing:

- Allow users to share match highlights or player stats on social media.

Out of Scope

Will-not-have Features:

Live Audio or Video Commentary:

- The project will not include live audio or video commentary, as the focus is on player stats and real-time data.

Player or Match Prediction:

- Predicting future match outcomes or player injuries will not be included in the initial scope.

Customizable User Interfaces for Fans:

- Advanced customization options for casual users (fans) to change the interface or select different visual themes will not be part of the first release.

Multiple Sport Integration:

- The app will be specific to soccer and will not support other sports initially.

Assumptions and Limitations

- **Data Accuracy:** The accuracy of player tracking and event detection will depend heavily on the quality of cameras and sensors used for the match. This may be limited by the available infrastructure.

- **Internet Connection:** Real-time data will require a stable and fast internet connection for proper streaming and data updates.

- **Live Match Availability:** The app's ability to track player performance depends on the availability of live match footage, which may not always be available.

- **Data Privacy:** The project will assume that player data shared for professional use (e.g., coaches, scouts) is legally collected and properly handled.

- **Compatibility:** The app and website will be designed to work on common devices (smartphones, tablets, desktops) but may not be optimized for every device or operating system in the initial version

4. High-level timeline

Phase	Description	Duration (weeks)	Deliverables
Research & Requirement Analysis	Research & Requirement Analysis Conduct a detailed review of existing soccer analysis technologies, gather requirements from professionals (coaches, scouts, players), and define the technical and functional needs for the app.	4	Research report, user requirements document, finalized technical specifications for app and computer vision integration.
Design & Planning	Design the system architecture, including the app's UI/UX, backend structure, and computer vision model.	3	System architecture diagrams, UI/UX wireframes, tech stack selection.
Implementation Part 1	Develop the core features: implement real-time data collection, and design basic player profiles with historical stats.	6	Working app prototype with live stats display, integrated computer vision for real-time tracking.

Implementation Part 2	Complete the integration of real-time match analysis, optimize the computer vision model, and refine app features such as player profile pages and stats history. Finalize the backend and database for performance data storage.	6	Fully integrated app, optimized player stats tracking, backend finalized, player profiles complete with historical data.
Testing & Evaluation	Perform system testing to ensure real-time data accuracy, evaluate the app's performance (speed, reliability), and test user interaction with the app (chatbot, profile viewing). Prepare final demo and documentation.	4	Tested app prototype, performance evaluation report, user feedback, final documentation.

5-Technology Stack & Theoretical Basis

Programming Languages:

- **Python:** Used in data analysis, AI model development (especially for computer vision and machine learning), and event detection. Python is highly suitable for data-heavy applications due to its extensive libraries (like TensorFlow, OpenCV).

- **JavaScript (React/Node.js):** React will be used for developing the frontend of the website (to display post-match stats and visualizations), while Node.js will handle backend services and real-time communication.

- **SQL (PostgreSQL or MongoDB):** For storing player stats, match data, and user information. PostgreSQL offers strong support for relational data, while MongoDB is great for handling unstructured data and scalability.

Frameworks:

- **React Native:** To build the mobile app, React Native allows for cross-platform mobile development (iOS and Android), saving time and resources by using a single codebase.
- **Flask/Django:** Lightweight frameworks for building the app's backend in Python. Flask can be used for simplicity and flexibility, while Django is preferred if needing a more structured framework with built-in features like authentication.
- **TensorFlow / OpenCV:** For computer vision and machine learning. TensorFlow will be used for model training and AI features, while OpenCV will handle image processing and player tracking.
- **Socket.IO / WebSockets:** For real-time communication between the server and app, allowing match stats to update live on the user interface.

APIs & Tools:

- **SportRadar API**: To pull real-time match data and player stats.
- **Twilio (for chatbot)**: To integrate the chatbot feature, making it easy for users to query match stats.
- **AWS / Google Cloud**: For cloud storage, scalable data processing, and hosting the app. Both provide high availability and computing power needed for processing large data during live matches.

Tools:

- **Postman:** For testing and ensuring that the APIs function correctly.
- **Git/GitHub:** For version control and team collaboration.

- ## Justification for Choices

- **Python**: It's the best choice for AI and computer vision due to its extensive libraries, such as TensorFlow for machine learning and OpenCV for image processing. Python's popularity in the data science field ensures strong community support and constant improvement in the tools and libraries we need.
- **React Native:** Since we are aiming for cross-platform mobile support (iOS and Android), React Native offers the advantage of a single codebase for both platforms, speeding up development and reducing maintenance costs. It's also great for building interactive UIs.

- **React and Node.js:** These are both modern and scalable technologies. React's component-based architecture allows for fast development and an efficient, reusable code structure. Node.js is fast, lightweight, and great for real-time applications, making it ideal for handling live match data.

- **TensorFlow / OpenCV:** TensorFlow offers advanced machine learning capabilities to train models that can predict player performance and events. OpenCV will be essential for real-time tracking and player identification through computer vision.

- **WebSockets:** Real-time updates are crucial for the live analysis feature of the app. WebSockets allow for a two-way communication channel, making it perfect for transmitting match data instantly to the app.

- **AWS/Google Cloud**: Both cloud platforms are well-suited for handling large-scale data and offer powerful computing resources to process and analyze match data in real-time. They also provide high security, uptime, and scalability to support growing user needs.

Program Focus

SWD (Software Development)

- **Algorithms:** Algorithms will be used for match event detection (e.g., identifying goals, assists, tackles) and for player tracking via computer vision (OpenCV).

- **Software Architecture:** The architecture follows a microservices approach, where different components of the system (e.g., live stats, backend, AI model, and frontend) work independently but communicate efficiently through APIs.

- **Systems Security:** Systems Security: User data and match statistics will be securely stored with proper encryption protocols. We will also implement OAuth2 for secure user authentication and ensure that all communication is encrypted (using HTTPS and SSL/TLS).

- ## DSAI (Data Science, AI, & Integration):

- **Data Collection & Analysis**: Player tracking data will be collected via cameras and sensors, processed by OpenCV, and used for generating real-time stats. Additionally, historical match data will be used to track player performance over time and build prediction models.

- **AI Models:** TensorFlow will be used to develop AI models for performance predictions (e.g., player ratings), as well as to assist with event detection and prediction based on historical data.

- **Lifecycle Management:** The data will go through a continuous feedback loop. As more match data is collected, it will refine the AI models for better accuracy. Real-time data will be processed immediately, and historical data will be stored for ongoing analysis and model training.

Success Metrics & Evaluation Plan

1. Performance (Speed, Latency, Accuracy)

Success Metric:

- **Latency**: The time taken for live match stats (e.g., goals, assists, player tracking) to update on the user interface. The system should process and display new data in real-time with **minimal delay** (under 3 seconds).

- **Speed**: The time it takes for the app to load player stats or match data after a user requests it. This should be under **2 seconds** for data retrieval.

- **Accuracy**: The precision with which the **computer vision system** identifies players and detects match events (goals, assists, fouls). A target accuracy rate of **95%** or higher will be aimed for.

Evaluation Plan:

- **Testing**: Perform load and stress testing using real match footage to evaluate system responsiveness. This will include tracking the time it takes for data to be processed and displayed during the game.

- **Benchmarking**: Compare performance against other similar sports analysis tools (e.g., Opta Sports or Stats Perform) in terms of real-time data updates and event detection accuracy.

2. Usability or User Satisfaction

Success Metric:

- **Usability**: The app and website should have an intuitive design, making it easy for users (especially coaches, scouts, and fans) to navigate through stats and player profiles.

- **User Satisfaction**: A **user satisfaction score** of at least **80%** (measured through surveys and feedback) will be considered a success.

Evaluation Plan:

- **Usability Testing**: Conduct usability testing with target users (coaches, scouts, players, and fans). Observe their interactions with the app and website to ensure ease of use and minimal friction.

- **Surveys**: After the app has been used, gather user feedback on their experience through **post-match surveys** and **user interviews**. The surveys will assess satisfaction with navigation, real-time data accuracy, and overall usability.

3. Reliability / Security

Success Metric:

- **System Reliability**: The system should be **available 99.9% of the time**, with no major downtimes during live matches.

- **Security**: All user data (especially player stats and personal data) must be securely stored and transmitted, with **no security breaches**.

Evaluation Plan:

- **Uptime Monitoring**: Use monitoring tools to ensure that the system remains available throughout the project lifecycle.

- **Security Testing**: Implement **data encryption** and secure **OAuth2 authentication** for users.

4. Data Quality / Model Accuracy (for DSAI)

Success Metric:

- **Data Quality**: The quality of data collected from the match (e.g., player position, events like goals) should be **accurate and reliable**. The system should **minimize errors** in event detection and player tracking.

- **Model Accuracy**: The AI model that predicts player performance and detects match events should have an **accuracy of 95%** or higher for event detection (goals, assists, etc.) and **80%** or higher for player performance predictions.

Evaluation Plan:

- **Model Testing**: Evaluate the AI models using standard **machine learning metrics** such as **precision**, **recall**, **F1-score**, and **accuracy**. Use a dataset of historical match data to test the model's prediction accuracy and event detection.

- **Data Quality Review**: Implement data validation checks to ensure that all input data (player movements, match events) is clean and reliable. Cross-check with manual annotations for accuracy.

5. Benchmarks and Overall Success Criteria

- **Benchmarking**: Compare the app's features against leading sports analytics platforms like **Opta Sports**, **Stats Perform**, or **FIFA's own tracking system** to assess how well it performs in real-time analysis and player tracking.

- **Overall Success**: If the system meets the following, it will be considered a success:

    - **Real-time performance** is smooth with minimal latency and accurate event tracking.

    - **User satisfaction** is above 80%, with positive feedback from coaches, players, and fans.

    - The system remains **reliable** with **99.9% uptime** and robust security protocols.

    - The **AI model** is effective with high prediction accuracy, and the **data collected** is clean and trustworthy.

Conclusion

To ensure that the project is successful, it will be evaluated using a combination of performance testing, user feedback, security checks, and data quality assessments. Regular **testing phases** will ensure that the system meets the specified metrics, and any issues will be resolved during the development process.

Team Roles & Responsibilities

Team Member	Program	Primary Role	Technical Contribution
Mohamed Wael - Ahmed Amr	DSAI Department	Al Model Developers .	- Lead the design, training, and optimization of the deep learning model for player tracking and event detection. -Oversee the dataset collection, preprocessing, and augmentation to ensure model robustness and accuracy. -Coordinate team progress and ensure milestones are met according to the timeline.

			-Monitor model performance, accuracy, and speed during the operation. -Assist in the collection, cleaning, and balancing of match data for model training..
			Develop machine learning models for event detection, player stats prediction, and match analysis using TensorFlow Conduct model evaluation with performance metrics like accuracy, precision, recall, and F1-score. Monitor and optimize model performance for higher speed and accuracy during real-time analysis.
Ahmed Sameh - Hesham Ashraf	SWD Department	Front/Backend developers and System integration	Develop interactive UI for displaying real-time player stats and match analysis using React. Implement data visualization components using Chart.js or D3.js for stats like goals, assists, and shots. Ensure a user-friendly design that accommodates

			real-time data and post-match stats.
			Improve model response time and help optimize app performance for quick, real-time updates. Collaborate on integrating player profiles to display stats for the last 5 matches for EX in a seamless way. Handle cloud storage and database management (PostgreSQL or MongoDB) to store player stats and historical data. Collaborate on integrating Al outputs into the mobile app for displaying real-time stats and event detection.
			Develop the backend architecture using Node.js or (Flask/Django) to process and store match data. Collaborate on integrating player profiles to display stats for the last 5 for EX matches in a seamless way. Ensure real-time data updates using WebSockets to display live match stats and player information.

			Ensure secure communication between the app, backend, and third-party APIs (e.g., SportRadar API). Design the database schema to track match data, player stats, and real-time events. . Collaborate on integrating Al outputs into the mobile app for displaying real-time stats and event detection.

Communication Plan

Meeting frequency:

- Weekly team meetings will be held to discuss progress, challenges, and upcoming tasks.
- weekly meetings with the project advisor will be scheduled to review milestones and receive feedback.
- Urgent meetings may be arranged when urgent technical or design issues arise.

Communication Tools:

- **Google meet/ Zoom:** for virtual meetings and advisor check-ins.
- **WhatsApp Group:** for daily communication, quick updates, and coordination among team members.
- **Email:** for formal correspondence and advisor communication.

Documentation and Collaboration Platforms:

- **Google Drive:** for storing project files, shared documents, datasets, and meeting notes.

- **GitHub:** for version control, code sharing, and collaborative development of the AI model and mobile application.

- **Trello:** for task management, tracking project progress, assigning responsibilities, and ensuring all milestones are completed on schedule.

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